Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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Background

2017

Ling et al

Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems

2020

Brown et al

Language Models are Few-Shot Learners

2021

Cobbe et al

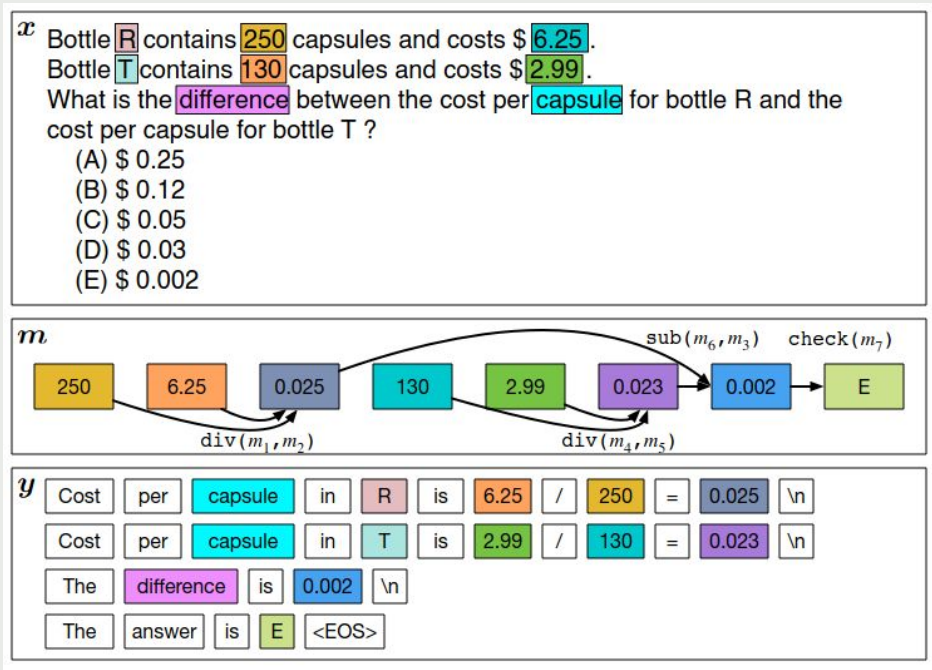
Training Verifiers to Solve Math Word Problems

Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems

July 2017, by Ling et al https://aclanthology.org/P17-1015/

Built a seq2seq LSTM-based model to break word problems down step by step. They achieved an accuracy of 36.4% when tested on a set of multiple-choice word problems, while the baseline models achieved around 20% (essentially random chance).

The model performed poorly by today's standards, but was an improvement over previous approaches, and this was done *before the advent of LLMs*. Transformers were invented month before, and GPT-1 wouldn't be released until the next year.



Left: example of the rationale provided by the model for solutions to math word problems.

Language Models are Few-Shot Learners

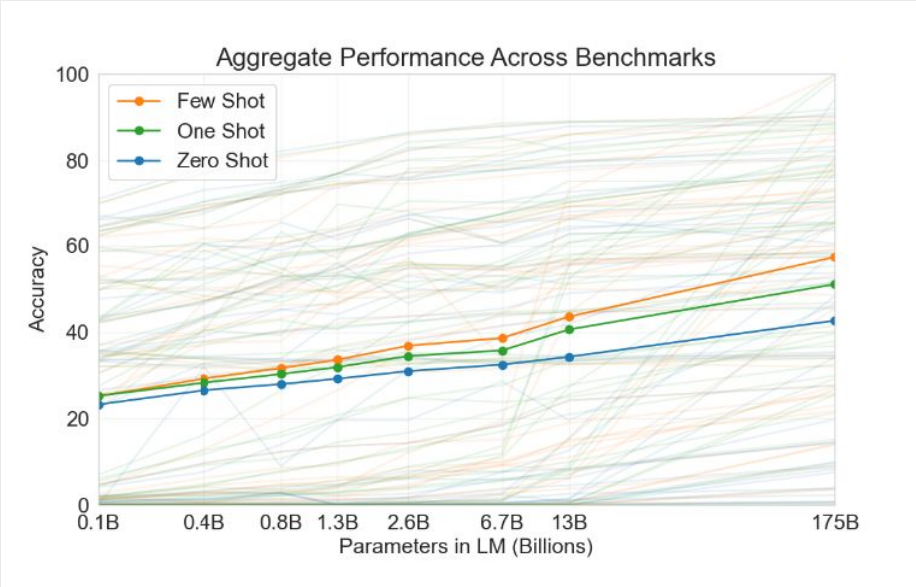
May 2020, by Brown et al https://arxiv.org/abs/2005.14165

Showed that LLMs are few-shot learners: you can get them to do something by giving a few examples in a prompt, rather than by fine-tuning. Model performance increases logarithmically with the number of parameters. That is, you get diminishing returns on performance as models get bigger. Few-shot performance is greater than one-shot, and both are greater than zero-shot, and this gap increases with model size (that is, the performance boost from few-shot vs zero-shot is greater with larger models), but all still scale logarithmically.

"**Few-Shot (FS)** - the model is given a few demonstrations of the task at inference time as conditioning [RWC+19], but no weights are updated. An example typically has a context and a desired completion (for example an English sentence and the French translation), and few-shot works by giving K examples of context and completion, and then one final example of context, with the model expected to provide the completion." (Brown et al, 6)

"**One-Shot (1S)** - similar to few-shot but with K = 1.' (Brown et al, 6)

"**Zero-Shot (0S)** - similar to few-shot but with a natural language description of the task instead of any examples." (Brown et al, 7)



Top right: performance of GPT-3 with few-shot, one-shot, and zero-shot prompts against several benchmarks.

Training Verifiers to Solve Math Word Problems

November 2021, by Cobbe et al https://arxiv.org/abs/2110.14168

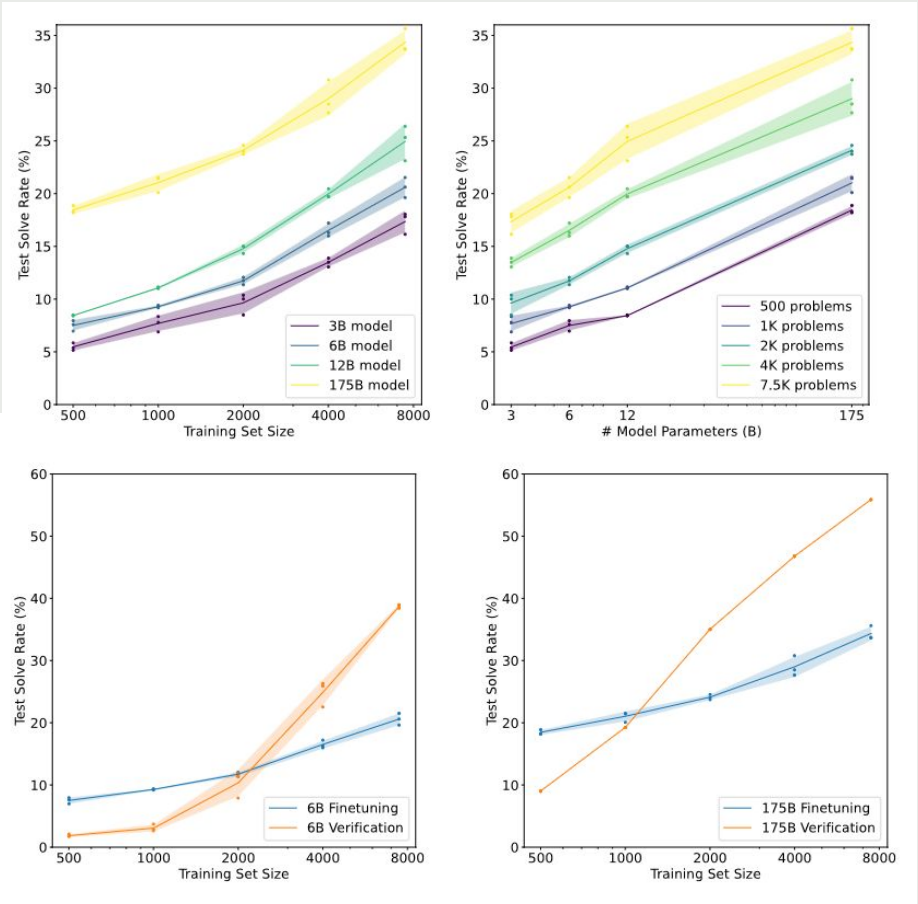
Showed you can fine-tune an LLM to solve math problems with chain-of-thought reasoning. They also released the GSM8K, "a dataset of 8.5K high quality problems at the grade school math level." (Cobbe et al, 2)

Fine-tuning: taking a model that was pre-trained on general data (e.g. an LLM) and further training it on data specific to the task.

Problem: Beth bakes 4, 2 dozen batches of cookies in a week. If these cookies are shared amongst 16 people equally, how many cookies does each person consume?
Solution: Beth bakes 4 2 dozen batches of cookies for a total of 4*2 = <<4*2=8>>8 dozen cookies
There are 12 cookies in a dozen and she makes 8 dozen cookies for a total of 12*8 = <<12*8=96>>96 cookies
She splits the 96 cookies equally amongst 16 people so they each eat 96/16 = <<96/16=6>>6 cookies
Final Answer: 6

Problem: Mrs. Lim milks her cows twice a day. Yesterday morning, she got 68 gallons of milk and in the evening, she got 82 gallons. This morning, she got 18 gallons fewer than she had yesterday morning. After selling some gallons of milk in the afternoon, Mrs. Lim has only 24 gallons left. How much was her revenue for the milk if each gallon costs $3.50?
Mrs. Lim got 68 gallons - 18 gallons = <<68-18=50>>50 gallons this morning.
So she was able to get a total of 68 gallons + 82 gallons + 50 gallons = <<68+82+50=200>>200 gallons.
She was able to sell 200 gallons - 24 gallons = <<200-24=176>>176 gallons.
Thus, her total revenue for the milk is $3.50/gallon x 176 gallons = $<<3.50*176=616>>616.
Final Answer: 616

Problem: Tina buys 3 12-packs of soda for a party. Including Tina, 6 people are at the party. Half of the people at the party have 3 sodas each, 2 of the people have 4, and 1 person has 5. How many sodas are left over when the party is over?
Solution: Tina buys 3 12-packs of soda, for 3*12= <<3*12=36>>36 sodas
6 people attend the party, so half of them is 6/2= <<6/2=3>>3 people
Each of those people drinks 3 sodas, so they drink 3*3=<<3*3=9>>9 sodas
Two people drink 4 sodas, which means they drink 2*4=<<4*2=8>>8 sodas
With one person drinking 5, that brings the total drank to 5+9+8+3= <<5+9+8+3=25>>25 sodas
As Tina started off with 36 sodas, that means there are 36-25=<<36-25=11>>11 sodas left
Final Answer: 11



Top right: performance of the fined-tuned GPT-3 model. Bottom right: performance with a verifier. Bottom left: sample of the GSM8K.

Recap

Cobbe et al
Showed you can fine-tune a pretrained model to reason about math problems.

Brown et al
Showed that LLMs have few-shot capabilities.

This paper (Wei et al) shows you can achieve the same result without training a model from scratch (that is, you can use an extant LLM) or fine-tuning it (that is, taking an extant LLM and training it further). PaLM has one-shot or few-shot capability for chain-of-thought prompting.

"Both of the above ideas, however, have key limitations. For rationale-augmented training and finetuning methods, it is costly to create a large set of high quality rationales, which is much more complicated than simple input–output pairs used in normal machine learning. For the traditional few-shot prompting method used in Brown et al. (2020), it works poorly on tasks that require reasoning abilities, and often does not improve substantially with increasing language model scale (Rae et al., 2021). In this paper, we combine the strengths of these two ideas in a way that avoids their limitations."

Wei et al, 2

Problem Areas

Arithmetic Reasoning

Commonsense Reasoning

Symbolic Reasoning

Arithmetic Reasoning - Method

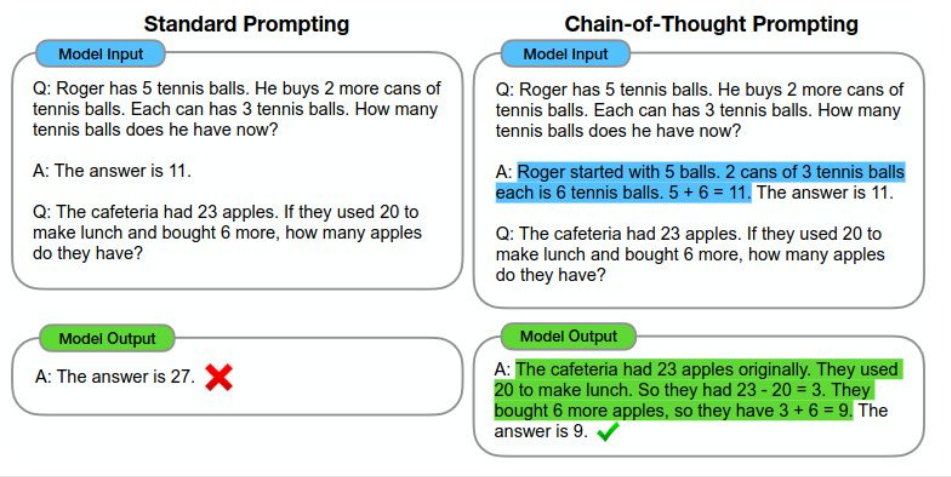
Ran 5 models:

1. GPT-3
2. LaMDA
3. PaLM
4. UL2 20B
5. Codex

against 5 math word problem benchmarks:

1. GSM8K
2. SVAMP
3. ASDiv
4. AQuA
5. MAWPS

Comparing standard 1-shot prompting vs chain-of-thought prompting.



Top right: examples of standard 1-shot prompting and chain-of-thought prompting.

Arithmetic Reasoning - Results

1. Only larger models (more than 8B parameters) see benefits from CoT. Small models actually performed worse with it.
2. Only more complicated problems see benefits from CoT.
3. Large models (GPT-3 175B and PaLM 540B) with CoT performed roughly as well as or better than prior fine-tuned models.

Why does it work?

Some ideas:
It makes the LLM restate the word problem as an equation, then solve that equation. Authors tested a variation where, instead of full CoT, the model output the equation before the answer. This did not meaningfully improve performance relative to standard prompting.

It lets the model use more compute for harder problems. Authors tested a version where the model output a "sequence of dots (. . .) equal to the number of characters in the equation needed to solve the problem." Frankly I don't understand how this would be an improvement over the equation-only version anyhow. In any event it did not improve on the baseline.

The prompt itself is allowing the model to more effectively use knowledge from pretraining. That is, the model does better not because it itself reasons, but because it has an example of reasoning which puts it in the right state of mind.
Authors tested a version where the CoT was given after the answer. This did not improve on the baseline, implying that the chain of reasoning must come before the answer in order to affect the accuracy.

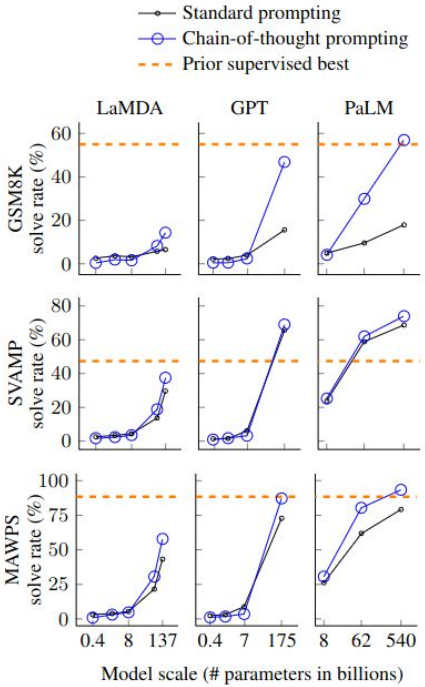


Figure 4: Chain-of-thought prompting enables large language models to solve challenging math problems. Notably, chain-of-thought reasoning is an emergent ability of increasing model scale. Prior best numbers are from Cobbe et al. (2021) for GSM8K, Jie et al. (2022) for SVAMP, and Lan et al. (2021) for MAWPS.

Right: comparisons of three models against three benchmarks, by model size.

Arithmetic Reasoning - How important is the example chain-of-thought?

The CoT prompts used in the benchmarks above were all written by an author of the paper referred to as Annotator A. The authors then redid benchmarks for LaMDA 137B on GSM8K and MAWPS, with prompts from

- Annotator A (the initial annotator)
- Annotator B
- Annotator C
- Annotator A again, but written more concisely

  For instance, whereas original [sic] chain of thought uses several short sentences (“’There were originally 9 computers. For each of 4 days, 5 more computers were added. So 5 * 4 = 20 computers were added. 9 + 20 is 29.”), the concise chain of thought would read “5 * 4 = 20 new computers were added. So there are 9 + 20 = 29 new computers in the server room now”. (p6)

- Alpha - a set of 8 exemplars from GSM8K
- Beta - a different set of 8 exemplars from GSM8K
- Gamma - a different set of 8 exemplars from GSM8K

All versions of CoT significantly outperformed standard prompting, but Annotator C really mogged Annotators A and B.

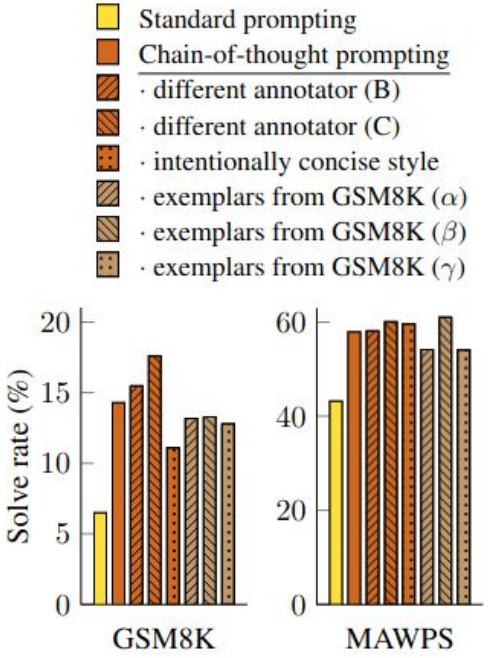


Figure 6: Chain-of-thought prompting has variance for different prompt examples (as expected) but outperforms standard prompting for various annotators as well as for different exemplars.

Right: comparisons of different prompt styles against two benchmarks.

Commonsense Reasoning

Method

Ran 5 models:

1. GPT-3
2. LaMDA
3. PaLM
4. UL2 20B
5. Codex

against 5 math word problem benchmarks:

1. CSQA - “commonsense questions about the world involving complex semantics that often require prior knowledge.”
2. StrategyQA - “requires models to infer a multi-hop strategy to answer questions.”
3. BIG-bench Date Understanding - “inferring a date from a given context.”
4. BIG-bench Sports Understanding - “determining whether a sentence relating to sports is plausible or implausible.”
5. SayCan - “mapping a natural language instruction to a sequence of robot actions from a discrete set.”

Comparing standard 1-shot prompting vs chain-of-thought prompting.

Results

- Increasing model size improved performance for all benchmarks
- CoT prompting yielded significant improvements for all benchmarks except CSQA

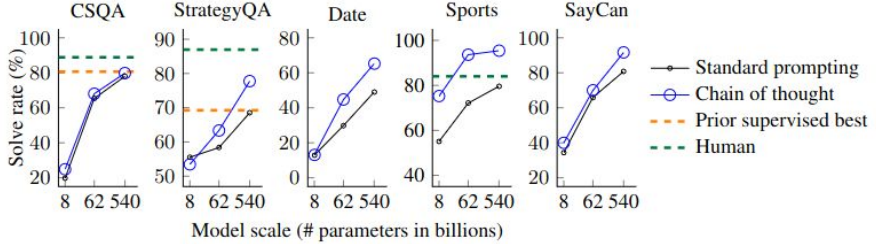


Figure 7: Chain-of-thought prompting also improves the commonsense reasoning abilities of language models. The language model shown here is PaLM. Prior best numbers are from the leaderboards of CSQA (Talmor et al., 2019) and StrategyQA (Geva et al., 2021) (single-model only, as of May 5, 2022). Additional results using various sizes of LaMDA, GPT-3, and PaLM are shown in Table 4.

Bottom right: PaLM performance with and without CoT prompting against five benchmarks.

Symbolic Reasoning

Method

“

Tasks. We use the following two toy tasks.

• **Last letter concatenation.** This task asks the model to concatenate the last letters of words in a name (e.g., “Amy Brown” → “yn”). It is a more challenging version of first letter concatenation, which language models can already perform without chain of thought.3 We generate full names by randomly concatenating names from the top one-thousand first and last names from name census data (https://namecensus.com/).

• **Coin flip.** This task asks the model to answer whether a coin is still heads up after people either flip or don’t flip the coin (e.g., “A coin is heads up. Phoebe flips the coin. Osvaldo does not flip the coin. Is the coin still heads up?” → “no”).

”

Wei et al, 8

Results

- ❖ CoT improves model performance in all examples
- ❖ CoT improves model performance in OOD problems (problems which are not in the training set), but increasing model size without CoT does not.
- ❖ This implies that CoT is especially good for generalizing to problems on which the model has no prior data.

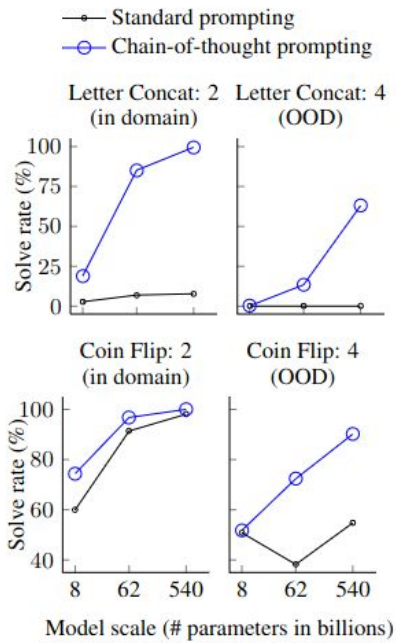


Figure 8: Using chain-of-thought prompting facilitates generalization to longer sequences in two symbolic reasoning tasks.

Bottom right: PaLM performance with and without CoT prompting for each of the tasks, with an in-domain and out-of-domain variety.

Summary

Models can learn to produce a chain of thought via few-shot prompting.

Models of a sufficient size show performance improvements in various subjects.

Models of sufficient size can generalize to problems which are similar to but more complicated than problems in the exemplars.

“Sufficient size” means having more than x billion parameters, where x is somewhere between 8 and 62.

The Paper's Influence

Large Language Models are Zero-Shot Reasoners, published May 2022 by Kojima et al, showed that you actually don't need to give models examples to induce CoT reasoning - simply insert the phrase "Let's think step by step," and the model will provide a chain of thought.

Solving math word problems with process- and outcome-based feedback, published November 2022 by Uesato et al, established a distinction between outcome-based and process-based approaches. In an outcome-based approach, a model is rewarded for having a correct final answer. In a process-based approach, the correctness of individual steps in the CoT matters.

Let's Verify Step by Step, published May 2023 by Lightman et al, iterated on Uesato et al, doing a comparison of outcome-based and process-based supervision, finding

> "1. We show that process supervision can train much more reliable reward models than outcome supervision. We use our state-of-the-art PRM to solve 78.2% of problems from a representative subset of the MATH test set.
> 2. We show that a large reward model can reliably approximate human supervision for smaller reward models, and that it can be used to efficiently conduct large-scale data collection ablations.
> 3. We show that active learning leads to a 2.6× improvement in the data efficiency of process supervision.
> 4. We release our full process supervision dataset, PRM800K, to promote related research."

OpenAI o1, released on September 12, 2024* was the first reasoning model, i.e. a model which uses CoT by default, without user prompting.

* Do you think they postponed it a day?